

# Introduction to Data Science in Biology

## INBRE Summer Biostatistics Workshop – Day 1

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## Welcome to the INBRE Summer Biostatistics Workshop

A 3-week introduction to modern data science methods for biological research

- No prior R experience assumed
- Emphasis on hands-on learning
- Biological and biomedical applications throughout
- Focus on reproducible scientific workflows

# Course Goals

By the end of this workshop, you should be able to:

- Work comfortably in R and RStudio
- Import, organize, and visualize biological datasets
- Understand the foundations of statistical inference
- Fit and interpret common statistical models
- Analyze modern high-dimensional biological data
- Create reproducible analysis reports

# Course Structure

3 Weeks

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4 Hours Per Day

## Daily Structure

- Morning: Lecture and Discussion
  - Afternoon: Hands-on R Lab
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- Week 1: Foundations of R and Data Science
  - Week 2: Statistical Modeling
  - Week 3: Genomics and High-Dimensional Data

## Repository Link

<https://github.com/alexmcclain/INBRE-Bios-Summer-26>

- All lecture notes, lab files, and datasets are in one place
- Download the repository once before the first lab
- Everything you need is already organized inside it
- Materials will be added throughout the course as we go

## Important

Do not move files around inside the folder. The file paths in the lab documents depend on the folder structure being intact.

# Downloading the Repository

## Option 1 — Download as a ZIP (recommended)

- 1 Go to <https://github.com/alexmcclain/INBRE-Bios-Summer-26>
- 2 Click the green **Code** button
- 3 Click **Download ZIP**
- 4 Unzip it somewhere permanent (Desktop or Documents)

## Option 2 — Clone with Git (if you have Git installed)

```
git clone https://github.com/alexmcclain/INBRE-Bios-Summer-26
```

## Important

Put the folder somewhere permanent. If you move it later, the file paths in the lab documents will break.

# Topics We Will Cover

- R and RStudio
- Data wrangling
- Data visualization
- Probability
- Statistical inference
- Regression
- Logistic regression
- Survival analysis
- Mixed models
- Genomics data
- Differential expression
- Multiple testing
- PCA
- Penalized regression
- Reproducible reporting
- Scientific communication

# Expectations

## What I Expect From You

- Attend and participate actively
- Ask questions
- Work through coding exercises
- Be willing to make mistakes
- Stay engaged during labs

## What You Should Expect From Me

- A supportive learning environment
- Practical biological examples
- Clear explanations
- Hands-on instruction
- Help debugging and interpreting results



You are not expected to memorize  
everything.

The goal is to learn how to think about data, how to approach problems systematically, and how to build analytical workflows.

# What is Data Science?

Data Science = Statistics + Computing + Subject Matter Expertise

- Statistics helps us learn from variability
- Computing allows us to analyze large, complex datasets
- Domain expertise helps us ask meaningful scientific questions

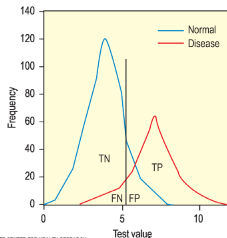
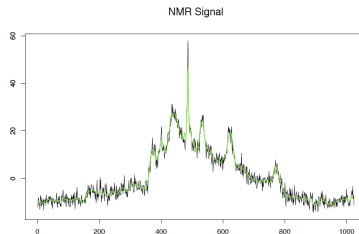
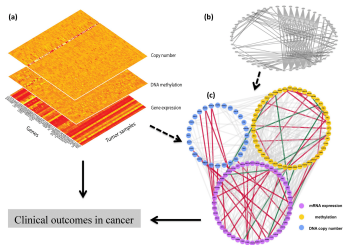
# Why Data Science Matters in Biology

Modern biology generates enormous amounts of data.

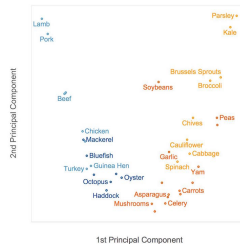
- Electronic health records
- Genomics and sequencing data
- Wearable sensors
- Environmental monitoring
- Medical imaging
- Single-cell technologies

Scientific discovery increasingly depends on data analysis.

# Examples of Biological Data



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# Examples of Scientific Questions

- Which genes are associated with disease?
- Does a treatment improve survival?
- Can biomarkers predict disease progression?
- How do environmental exposures affect health?
- Which patients are at highest risk?

## Key Idea

The scientific question should drive the analysis.

# Data Alone Are Not Enough

Having data is not the same thing as understanding data.

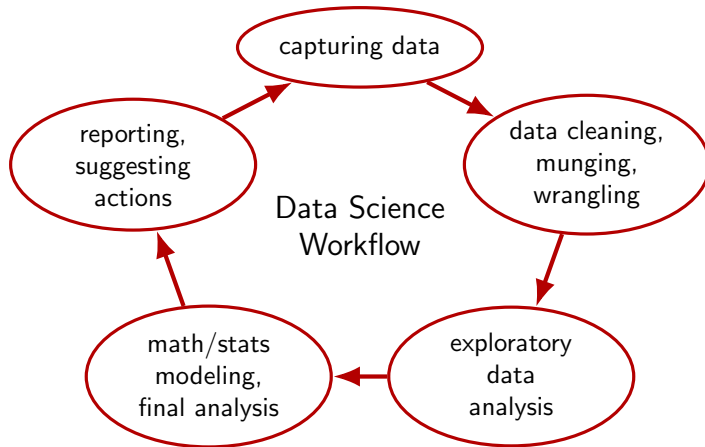
- Data can be incomplete
- Data can be noisy
- Data can be biased
- Data can be misleading
- Poor analysis can produce incorrect conclusions

Good science requires careful thinking.

# The Data Analysis Process

- 1 Ask a scientific question
- 2 Collect and organize data
- 3 Explore and visualize patterns
- 4 Fit statistical models
- 5 Interpret results
- 6 Communicate findings clearly

# Data Science Workflow





# What is Reproducibility?

## Reproducible Research

Someone else should be able to reproduce your analysis using your data and code.

- Organize files carefully
- Save analysis scripts
- Document decisions
- Keep raw data unchanged
- Avoid manual copy-paste workflows

# Why Reproducibility Matters

- Science builds on previous work
- Analyses must be transparent
- Results should be verifiable
- Reproducibility improves collaboration
- You will forget what you did six months from now

## Common Problem

“I got a result, but I do not remember how I produced it.”

# Good Habits for Reproducible Research

- Use scripts instead of clicking menus
- Organize projects consistently
- Name files clearly
- Save intermediate results carefully
- Comment your code
- Keep analyses readable

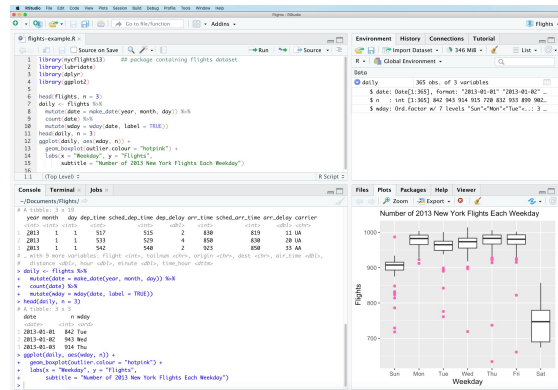
Future-you is one of your collaborators.

# What is R?

- R is a programming language designed for statistics and data analysis
- Widely used in:
  - biostatistics
  - epidemiology
  - genomics
  - bioinformatics
  - public health
- Free and open-source
- Large scientific community and ecosystem

# What is RStudio?

- RStudio is an integrated development environment (IDE) for R
- Makes it easier to:
  - write code
  - manage files
  - create figures
  - organize projects
  - build reports
- Today, we will learn how to navigate the RStudio interface and organize projects reproducibly.



RStudio Interface

# What Makes Someone Good at Data Science?

- Curiosity
- Persistence
- Attention to detail
- Clear scientific thinking
- Comfort with uncertainty
- Willingness to debug problems

Coding skill develops through practice.

You will get stuck.

That is normal.

Learning data science requires experimentation, mistakes, debugging, and repetition.

This afternoon we will:

- ① Install R and RStudio
- ② Learn the RStudio interface
- ③ Create projects and scripts
- ④ Discuss file organization
- ⑤ Practice reproducible workflows



# Questions?

Welcome to the workshop.